ignition of flammable vapours should not be permitted in these compartments except as permitted in 12.6.9.
12.6.8.3 Accumulator batteries should not be located in crew accommodation.
12.6.9 No electrical equipment should be installed in any space where flammable mixtures are liable to collect including those in compartments assigned principally to accumulator batteries, in paint lockers, acetylene stores or similar spaces, unless the

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SOURCE: MSC.36(63).pdf
flammable mixtures are liable to collect including those in compartments assigned principally to accumulator batteries, in paint lockers, acetylene stores or similar spaces, unless the Administration is satisfied that such equipment is:
12.6.10
.1 essential for operational purposes;
.2 of a type which will not ignite the mixture concerned;
.3 appropriate to the space concerned; and
.4 appropriately certified for safe usage in the dusts, vapours or gases likely to be encountered.
The following additional requirements from .1 to .7 should be met, and requirements from .8 to .13 should be met . also for non-metallic craft:
.1 The electrical distribution voltages throughout the craft may be either direct current or alternating current and should not exceed:
.1.1
.1.2
500 V for power, cooking, heating and other permanently connected equipment; and ·
250 V for lighting, internal communications and receptacle outlets.
The Administration may accept higher voltage for propulsion purposes .
. 2 For electrical power distribution, two-wire, three-wire or four-wire insulated systems should be used. Where applicable,

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SOURCE: MSC.36(63).pdf
receptacle outlets.
The Administration may accept higher voltage for propulsion purposes .
. 2 For electrical power distribution, two-wire, three-wire or four-wire insulated systems should be used. Where applicable, the requirements of 7.5.6.4 or 7.5.6.5 should also be met .
. 3 Effective means should be provided so that voltage may be cut off from each and every circuit and sub-circuit and from all apparatus as may be necessary to prevent danger .
. 4 Electrical equipment should be so designed that the possibility of accidentally touching live parts, rotating or moving parts as well as heated surfaces which might cause burns or initiate fire is minimized.

.5 Electrical equipment should be adequately secured. The probability of fire or dangerous consequences arising from damage to electrical equipment should be reduced to an acceptable. minimum .
. 6 The rating or appropriate setting of the overload · protective device for each circuit should be permanently indicated at the

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SOURCE: MSC.36(63).pdf
damage to electrical equipment should be reduced to an acceptable. minimum .
. 6 The rating or appropriate setting of the overload · protective device for each circuit should be permanently indicated at the location of the protection device .
. 7 Where it is impracticable to provide electrical protective devices for certain cables supplied from batteries, e.g. within battery compartments and in engine starting circuits, unprotected cable runs should be kept as short as possible and special precautions should be taken to minimize risk of faults, e.g. use of single core cables with additional sleeve over the insulation of each core, with shrouded terminals .
. 8 In order to minimize the risk of fire, structural damage, ele~trical shock and radio interference due to lightning strike or electrostatic discharge, all metal parts of the craft should be bonded together, in so far as possible in consideration of galvanic corrosion between dissimilar metals, to form a continuous electrical system, suitable for the earth return of electrical equipment and to connect the craft to the water when water-b6rn. The bonding of isolated components

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SOURCE: MSC.36(63).pdf
to form a continuous electrical system, suitable for the earth return of electrical equipment and to connect the craft to the water when water-b6rn. The bonding of isolated components inside the structure is not generally necessary, except in fuel tanks .
. 9 Each pressure refuelling point should be provided with a means of bonding the fuelling equipment to the craft .
. 10 Metallic pipes capable of generating electrostatic discharges, due to the flow of liquids and gases should be bonded so as to be electrically continuous throughout their length and should be adequately earthed .
. 11 Primary conductors provided for lightning discharge currents

should have a minimum cross section of 50 mm 2 in copper or equivalent surge carrying capacity in aluminium .
. 12 Secondary conductors provided for the equalization of static discharges, bonding of equipment, etc., but not for carrying lightning discharges should have a fllinimum cross section of 5 mm 2 copper or equivalent surge current carrying capacity in aluminiwn .
13 The electrical resistance between bonded objects and the basic structure should not exceed 0.05 Ohms except where it can be

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SOURCE: MSC.36(63).pdf
5 mm 2 copper or equivalent surge current carrying capacity in aluminiwn .
13 The electrical resistance between bonded objects and the basic structure should not exceed 0.05 Ohms except where it can be demonstrated that a higher resistance will not cause a hazard. The bonding path should have sufficient cross-sectional area to carry the maximum current likely to be imposed on it without excessive voltage drop.
W/1687n/pp/EWP
(
RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
ADOPTION OF THE INTERNATIONAL CODE OF SAFETY FOR HIGH SPEED CRAFT 12.7
12.7.1
12.7.2
12.7.3
General
MSC 62123/Add.2
ANNEX 18
Page 119
PART B - REQUIREMENTS FOR PASSENGER CRAFT
Separation and duplication of electrical supply should be provided for duplicated consumers of essential services. During normal operation the systems may be connected to the same power-bus, but facilities for easy separation should be provided. Each system should be able to supply all equipment necessary to maintain the control of propulsion, steering, stabilization, navigation, lighting and ventilation, and allow starting of the largest essential

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SOURCE: MSC.36(63).pdf
should be able to supply all equipment necessary to maintain the control of propulsion, steering, stabilization, navigation, lighting and ventilation, and allow starting of the largest essential electric motor at any load. Automatic load-dependent disconnection of non-essential consumers may be allowed.
Emergency source of electrical power
Where the main source of electrical power is located in two or more compartments which are not contiguous, each of which has its own

self-contained systems, including power distribution and control systems, completely independent of each other and such that a fire or other casualty in anyone of the spaces will not affect the power distribution from the others, or to the services required by 12.7.3 or 12.7.4, the requirements of 12.3.1, 12.3.2 and 12.3.4 may be considered satisfied without an additional emergency source of electrical power, provided that:
.1 there is at least one generating set, meeting the requirements of 12.3.12 and of sufficient capacity to meet the requirements of 12.7.3 or 12.7.4 in each of at least two non-contiguous spaces;
.2 the arrangements required by .1 in each such space are

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SOURCE: MSC.36(63).pdf
of 12.3.12 and of sufficient capacity to meet the requirements of 12.7.3 or 12.7.4 in each of at least two non-contiguous spaces;
.2 the arrangements required by .1 in each such space are equivalent to those required by 12.3.6.1, 12.3.7 to 12.3.11 and 12.4 so that a source of electrical power is available at all times to the services required by 12.7.3 or 12.7.4; and